Classical Mechanics

Lecture 6: Hamiltonian Mechanics and Phase-Space Flow

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Chapter 1

Hamiltonian Mechanics and Phase-Space Flow

Tonight we turn to Hamiltonian mechanics. This is still classical mechanics, not quantum mechanics, but the change of language will matter enormously when we reach quantum theory. The immediate purpose is more elementary: we want a description in which a complete state is a point in phase space and the law of motion carries that point along a reversible flow.

The route has three stages. First we ask what information a state must contain. Then a Legendre transformation replaces velocities by canonical momenta. Finally Hamilton's equations turn one function, the Hamiltonian, into the velocity field of the entire phase space.

1.1 Reversible Laws and Complete States

Begin with a coin whose state is either heads or tails. Imagine that time advances in discrete steps. One possible rule leaves the coin unchanged: heads goes to heads and tails goes to tails. Another rule flips it at every step. In either case the present determines not only the next state but also the preceding one. The evolution can be run forward or backward without ambiguity.

For a system with many possible states, represent each state by a point and draw an arrow from every point to its successor. A reversible rule permits exactly one arrow into and one arrow out of every point. For a finite state space, the arrows must therefore decompose into cycles. If the cycles are marked by different labels, such as $+1$ and -1, the label is conserved: a trajectory cannot jump from one cycle to another.

Now contrast this with a many-to-one rule. Several states may have the same successor. Forward prediction remains possible, but retrodiction does not: after reaching the common image, we cannot infer which predecessor was occupied. Distinguishable histories have been erased.

The same idea can be pictured by covering the state space with red dots and letting every dot move. Under a reversible rule no dot disappears, splits, or merges with another. In a continuous Hamiltonian system something stronger is true: phase-space volume is preserved. The points move as an incompressible fluid. Reversibility by itself does not prove incompressibility for every conceivable continuous flow; the canonical structure of Hamilton's equations does. We shall see the local reason for that structure in the crossed derivatives of those equations.

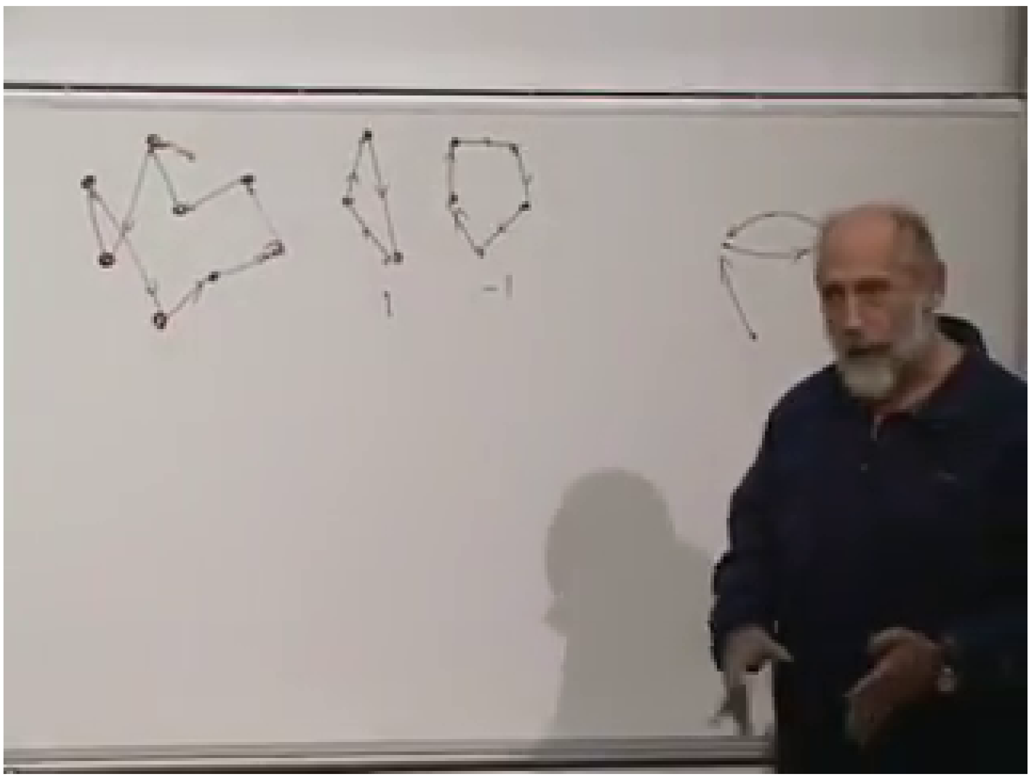

Figure 1.1: Discrete evolution at 00:08:05. The central cycles belong to separate sectors, while the map at the left illustrates how several distinct histories can merge when backward evolution is not unique.

The phrase *phase space* will mean the space of complete instantaneous states. A point in it must carry enough information for the equations of motion to determine a unique trajectory. If the microscopic law is reversible, the same point also determines the past.

1.2 Why Position Is Not a State

Suppose a particle is observed at a position q. Where will it be one second later? Position alone does not answer the question. The particle may be moving in any direction and at any speed. A complete state must include a second set of data,

$$(q_i, \dot{q}_i) \qquad \text{or, after a change of variables,} \qquad (q_i, p_i). \tag{1.1}$$

The coordinates q_i form configuration space. The pairs (q_i, p_i) form phase space.

This distinction is already visible in Newton's law. For N Cartesian coordinates,

$$m\ddot{x}_i = F_i, \qquad i = 1, \dots, N, \tag{1.2}$$

is a set of N second-order equations. Define

$$p_i = m\dot{x}_i. \tag{1.3}$$

The same physics becomes $2N$ first-order equations,

$$\boxed{\dot{x}_i = \frac{p_i}{m}, \qquad \dot{p}_i = F_i.} \tag{1.4}$$

This rewriting is not yet the full Hamiltonian construction. Any ordinary second-order equation can be made first order by introducing an extra variable. Hamiltonian mechanics demands the more special pairing of coordinates and canonical momenta, with both first-order equations generated by derivatives of a single function.

Question from the audience. Why reformulate $F = ma$ if it already predicts the motion?

Response. The Lagrangian formulation freed us from Cartesian coordinates. Motion on a plane may be described by x and y, while a pendulum or a particle on a sphere is more naturally described by angles. Hamilton's formulation reveals an additional structure: mechanics becomes a flow on a space of canonical pairs. The reformulation is useful in classical mechanics and becomes indispensable in quantum mechanics. Historical motives are harder to reconstruct than the mathematics, so the important test is what the construction accomplishes.

Imagine placing one identical system at every phase-space point. Each copy has an instantaneous velocity $(\dot{q}_i, \dot{p}_i)$, so the equations define a vector field on phase space. Following that field carries the whole cloud of systems forward. Our task is to find the function that generates this flow.

1.3 The Legendre Transformation

The required mathematical device is a Legendre transformation. Start with one variable V and a differentiable function $L(V)$. Define the conjugate variable

$$P = \frac{dL}{dV}. \tag{1.5}$$

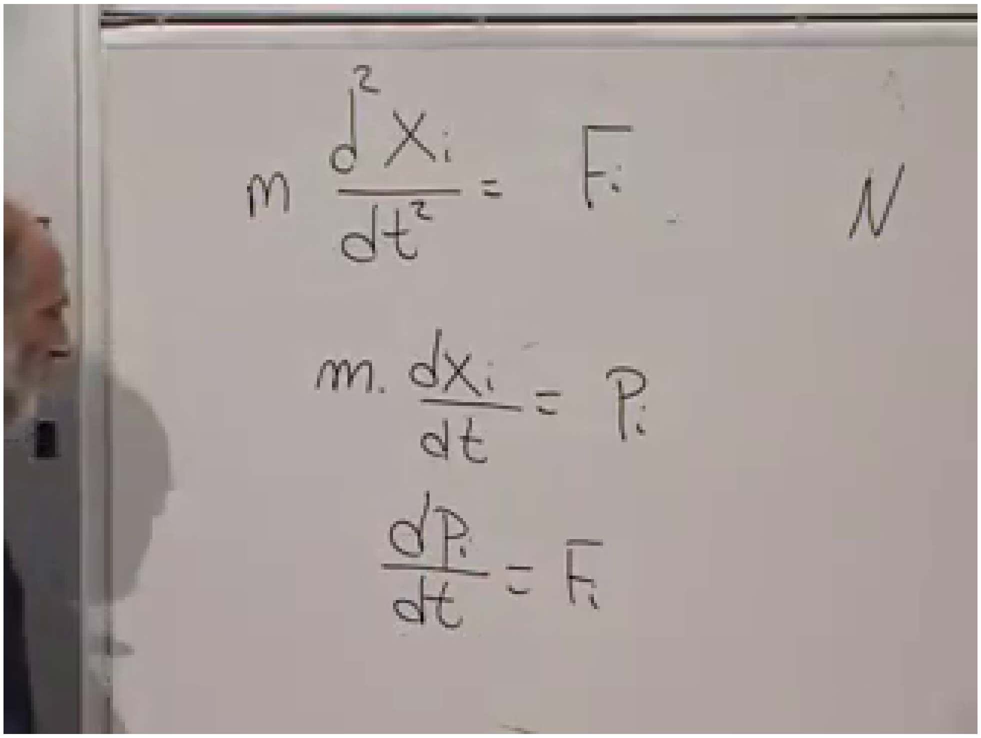

Figure 1.2: The Newton equation rewritten as two first-order equations at 00:18:48. No new physics has been added; the complete state has been made explicit.

Assume, at least in the region of interest, that this relation can be inverted to give $V = V(P)$. Geometrically the graph of $P(V)$ must not double back in a way that assigns several relevant values of V to the same P.

With convenient additive constants one may write

$$L(V) = \int_0^V P(V')\,dV', \qquad H(P) = \int_0^P V(P')\,dP'. \tag{1.6}$$

For a monotone graph through the chosen origin, these are complementary areas inside a rectangle of area PV:

$$H(P) + L(V) = PV. \tag{1.7}$$

The invariant definition, independent of the choice of additive constants, is

$$\boxed{H(P) = PV - L(V),} \qquad P = L'(V), \tag{1.8}$$

where V on the right is to be expressed as a function of P.

The derivative of the transformed function follows by varying Eq. (1.8):

$$\begin{aligned} \delta H &= P\,\delta V + V\,\delta P - \frac{dL}{dV}\,\delta V \\ &= P\,\delta V + V\,\delta P - P\,\delta V = V\,\delta P. \end{aligned} \tag{1.9}$$

Therefore

$$\boxed{\frac{dH}{dP} = V.} \tag{1.10}$$

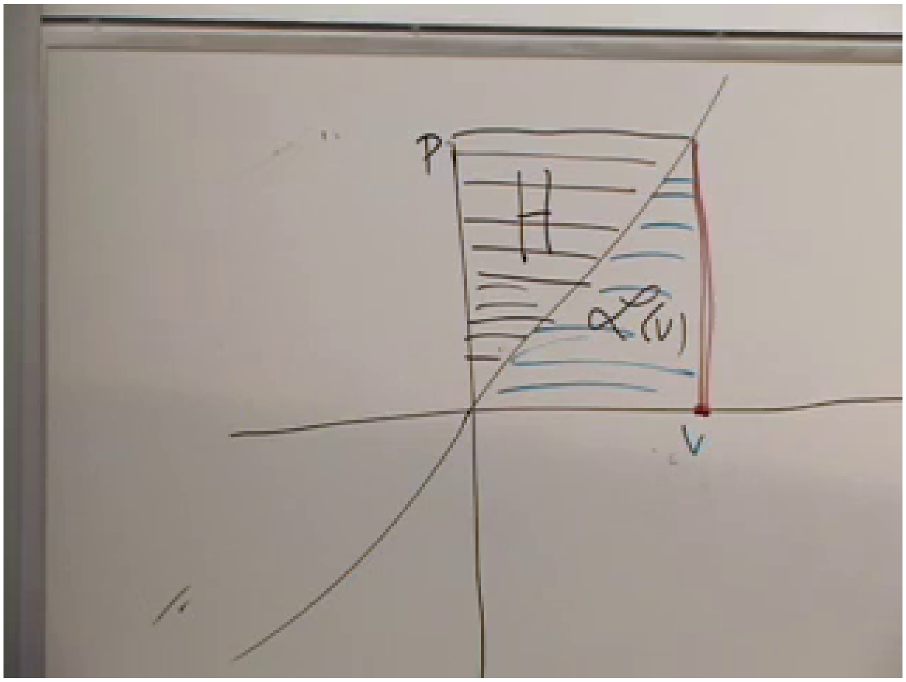

Figure 1.3: The complementary-area construction at 00:32:05. The area labeled $L(V)$ lies under $P(V)$; the remaining part of the PV rectangle is the transformed function $H(P)$.

Differentiation created P from $L(V)$; after the transform, differentiation creates V from $H(P)$. This reciprocity is the point of the construction.

For several variables, let $L = L(v_1, \ldots, v_N)$ and define

$$p_i = \frac{\partial L}{\partial v_i}. \tag{1.11}$$

If these equations can be inverted for all the v_i, the transform is

$$H(p_1, \ldots, p_N) = \sum_i p_i v_i - L, \tag{1.12}$$

and the same cancellation gives

$$\frac{\partial H}{\partial p_i} = v_i. \tag{1.13}$$

Legendre transformations appear repeatedly in thermodynamics, where changing which variables are held fixed produces the familiar thermodynamic potentials. Here we use the same mathematics to exchange velocities for momenta.

1.3.1 Coordinates Go Along for the Ride

In mechanics the Lagrangian is not merely $L(v)$, but

$$L = L(q, v, t), \qquad v_i = \dot{q}_i. \tag{1.14}$$

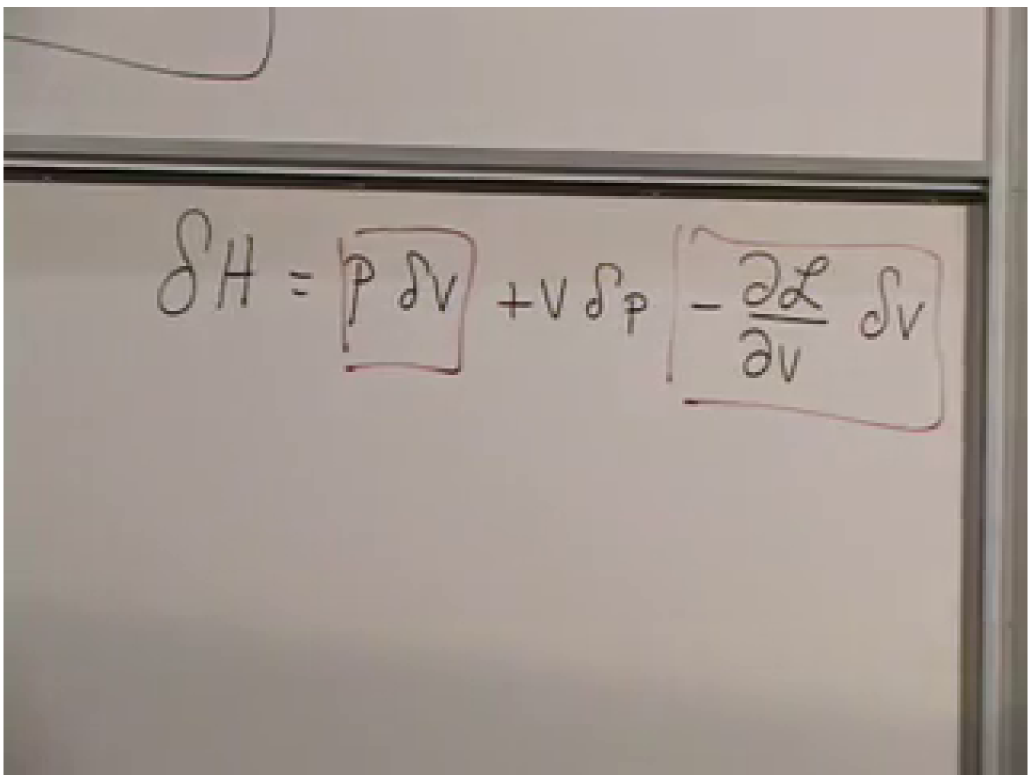

Figure 1.4: The cancellation in the variation of $H = PV - L$ at 00:36:30. The boxed terms cancel because $dL/dV = P$, leaving $\delta H = V\,\delta P$.

The transformation is performed only in the velocity variables. During it, the coordinates q_i and time t are held fixed as parameters:

$$p_i = \frac{\partial L}{\partial v_i}, \qquad H(q,p,t) = \sum_i p_i v_i - L(q,v,t). \tag{1.15}$$

The inverted velocities $v_i = v_i(q,p,t)$ are then substituted on the right, so H is genuinely a function of q, p, t.

Question from the audience. May the velocity determined by the momentum also depend on position?

Response. Yes. The Legendre transformation holds q fixed, but the relation $p_i = \partial L/\partial v_i$ may contain q. A particle in curvilinear coordinates is a standard example: the coefficients in the kinetic energy depend on position, so solving for v_i generally produces $v_i(q,p)$.

Local invertibility is controlled by the velocity Hessian

$$W_{ij} = \frac{\partial^2 L}{\partial v_i \partial v_j}. \tag{1.16}$$

For ordinary unconstrained mechanics, $\det W \neq 0$ allows the velocities to be solved locally in terms of the momenta. A singular Hessian does not mean that nature is inconsistent; it signals a constrained system requiring a more careful Hamiltonian treatment.

1.4 Hamilton's Equations

Now vary the mechanical transform while allowing both q_i and p_i to change. At fixed time,

$$\begin{aligned} \delta H &= \sum_i (p_i\,\delta v_i + v_i\,\delta p_i) - \sum_i \frac{\partial L}{\partial q_i}\,\delta q_i - \sum_i \frac{\partial L}{\partial v_i}\,\delta v_i \\ &= \sum_i v_i\,\delta p_i - \sum_i \frac{\partial L}{\partial q_i}\,\delta q_i. \end{aligned} \tag{1.17}$$

The two terms proportional to δv_i cancel because $p_i = \partial L/\partial v_i$. Reading off the coefficients gives

$$\frac{\partial H}{\partial p_i} = v_i = \dot{q}_i, \qquad \frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i}. \tag{1.18}$$

The Euler–Lagrange equations say

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} = \frac{\partial L}{\partial q_i}. \tag{1.19}$$

Since the derivative on the left is $\dot{p}_i$, we arrive at

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i}.} \tag{1.20}$$

The equations are first order and nearly symmetric in q_i and p_i. Their derivatives are crossed: the rate of change of q_i comes from a momentum derivative, while the rate of change of p_i comes from a coordinate derivative. The minus sign is essential. It makes the phase-space velocity field

$$\boldsymbol{u}_H = \left(\frac{\partial H}{\partial p_1}, \ldots, \frac{\partial H}{\partial p_N}, -\frac{\partial H}{\partial q_1}, \ldots, -\frac{\partial H}{\partial q_N}\right) \tag{1.21}$$

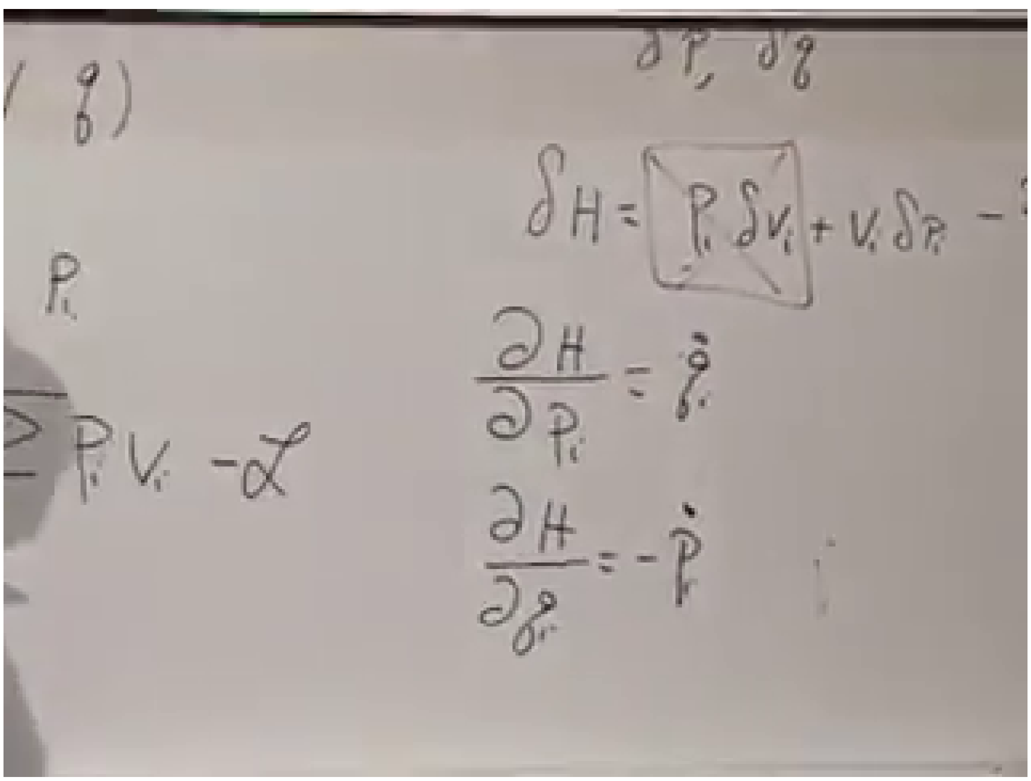

Figure 1.5: The completed pair of Hamilton equations at 00:52:40. The crossed derivatives and the single minus sign encode the canonical phase-space structure.

divergence-free:

$$\nabla_{q,p} \cdot \boldsymbol{u}_H = \sum_i \left(\frac{\partial^2 H}{\partial q_i \partial p_i} - \frac{\partial^2 H}{\partial p_i \partial q_i} \right) = 0. \tag{1.22}$$

Thus the red-dot fluid is locally incompressible. This is the elementary coordinate form of Liouville's theorem.

Question from the audience. Is Hamiltonian mechanics merely the trick of replacing each second-order equation by two first-order equations?

Response. No. Doubling the variables is always possible, but an arbitrary doubled system need not have the crossed-gradient form in Eq. (1.20). The canonical form, with one Hamiltonian generating both equations, is the stronger statement.

Question from the audience. What happens if the relation between velocity and momentum cannot be inverted?

Response. For the regular systems being discussed, the ordinary Legendre transform then fails and $H(q,p)$ cannot be constructed by this recipe. Such a model may be pathological, but singular Lagrangians also arise in legitimate constrained and gauge systems. They require additional constraint equations rather than the naive transform.

Question from the audience. Does friction, including terminal velocity, violate the Hamiltonian picture?

Response. A friction law for a reduced set of variables is generally not Hamiltonian: phase-space volumes contract and mechanical information is discarded. At the microscopic level, however, the object and its environment may form a larger Hamiltonian system. Energy and information then flow into unobserved molecular, acoustic, or radiative degrees of freedom.

Question from the audience. Where, exactly, did the Legendre transformation enter the mechanics?

Response. It is the passage from $L(q,v)$ to $H(q,p) = \sum_i p_i v_i - L$, with $p_i = \partial L/\partial v_i$ and the velocities eliminated in favor of the momenta. The geometric area picture explains the reciprocal derivatives; the cancellation in Eq. (1.17) is what turns that reciprocity into Hamilton's equations.

Question from the audience. Can a Legendre transform fail because the velocity becomes singular or several velocities correspond to one momentum?

Response. Yes. Global one-to-one behavior can fail even if a local branch exists, and the Hessian can become singular at special points. Before using the ordinary Hamiltonian construction one must specify a region where the momentum–velocity relation is regular, or adopt the constrained formalism appropriate to the system.

Question from the audience. Do infinitely many degrees of freedom, as in a field, automatically produce friction?

Response. No. A classical field can have infinitely many canonical pairs and still evolve Hamiltonianly. Apparent damping arises when energy is carried away into modes that are omitted from the reduced description. An accelerating charge, for example, may lose mechanical energy to radiation, while the combined particle-plus-field system retains the full accounting.

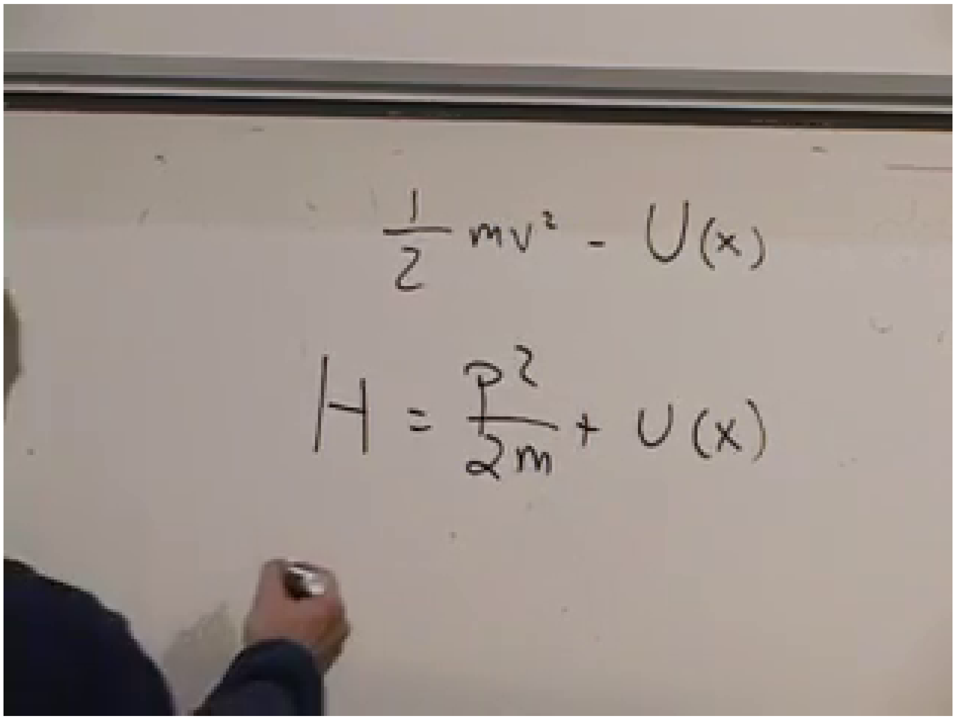

Figure 1.6: The ordinary-particle Hamiltonian at 01:12:35. The velocity has been eliminated, leaving $H = p^2/(2m) + U(x)$.

1.5 The Ordinary Particle

For one particle in a potential $U(x)$,

$$L(x, \dot{x}) = \frac{1}{2}m\dot{x}^2 - U(x). \tag{1.23}$$

The canonical momentum is

$$p = \frac{\partial L}{\partial \dot{x}} = m\dot{x}, \qquad \dot{x} = \frac{p}{m}. \tag{1.24}$$

Its Legendre transform is

$$\begin{aligned} H &= p\dot{x} - L \\ &= p\frac{p}{m} - \left(\frac{1}{2}m\frac{p^2}{m^2} - U(x)\right) \\ &= \boxed{\frac{p^2}{2m} + U(x)}. \end{aligned} \tag{1.25}$$

For this familiar Lagrangian, the Hamiltonian is the total energy $T + U$.

The first Hamilton equation gives

$$\dot{x} = \frac{\partial H}{\partial p} = \frac{p}{m}, \tag{1.26}$$

which is the momentum–velocity relation. The second gives

$$\dot{p} = -\frac{\partial H}{\partial x} = -\frac{dU}{dx}. \tag{1.27}$$

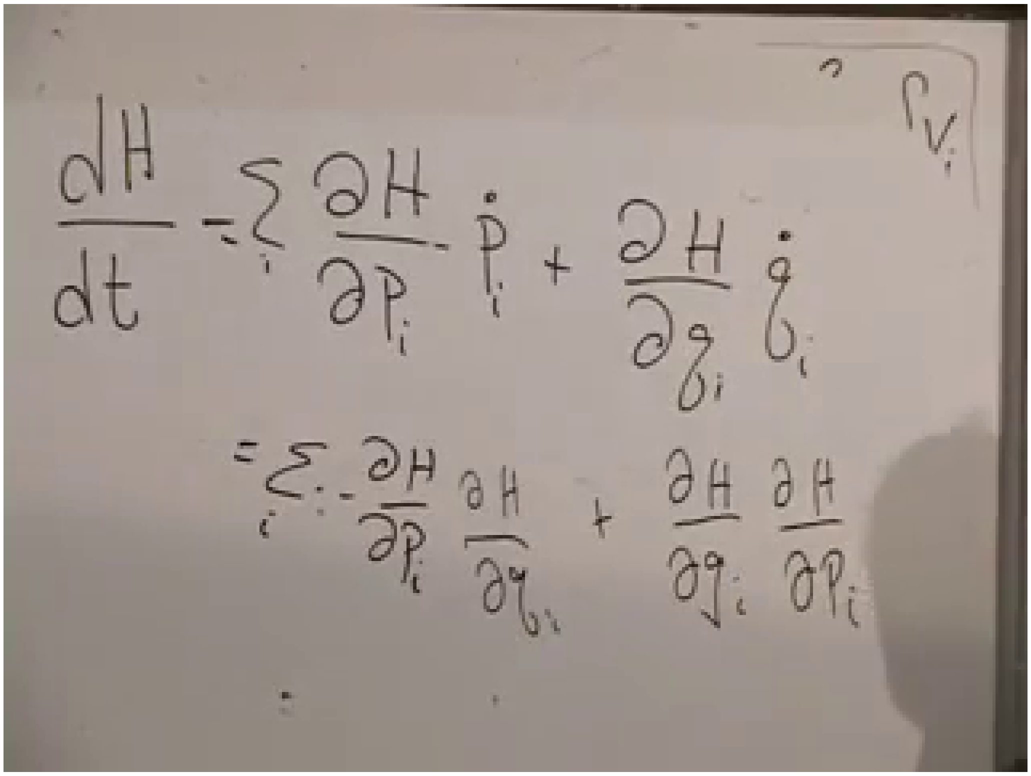

Figure 1.7: Energy conservation at 01:16:10. Substituting Hamilton's equations into dH/dt makes the two products cancel term by term.

Since $p = m\dot{x}$, this is Newton's equation

$$m\ddot{x} = -\frac{dU}{dx}. \tag{1.28}$$

The reformulation has changed the organization of the equations, not their physical predictions.

1.6 Energy and the Geometry of Its Contours

Suppose $H(q, p)$ has no explicit time dependence. Along a trajectory,

$$\begin{aligned} \frac{dH}{dt} &= \sum_i \left(\frac{\partial H}{\partial q_i} \dot{q}_i + \frac{\partial H}{\partial p_i} \dot{p}_i \right) \\ &= \sum_i \left(\frac{\partial H}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial H}{\partial p_i} \frac{\partial H}{\partial q_i} \right) = 0. \end{aligned} \tag{1.29}$$

The cancellation is pairwise, so

$$\boxed{H = \text{constant along the motion.}} \tag{1.30}$$

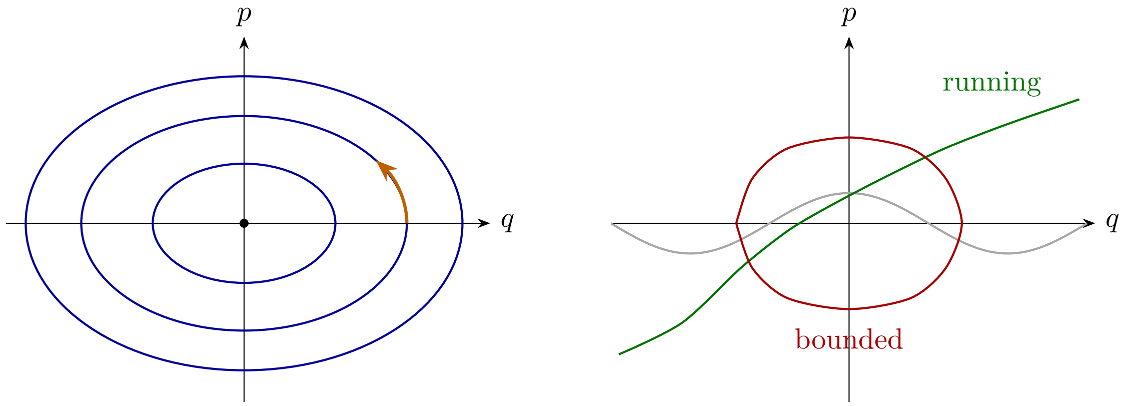

Figure 1.8: Typeset phase portraits. Small energy in a stable well produces a closed orbit. Above a periodic potential barrier, the coordinate can keep running and the constant-energy curve is open.

Question from the audience. Is the converse true? Does energy conservation by itself imply Hamiltonian dynamics?

Response. No. A time-independent Hamiltonian evolving by Hamilton's equations is conserved, but a dynamical rule may preserve some function called energy without possessing canonical Hamiltonian structure. Energy conservation is one consequence, not a complete characterization, of Hamiltonian flow.

For one degree of freedom, constant values of $H(q, p)$ are curves in the q-p plane. The phase point must move along the curve on which it starts. For a harmonic oscillator,

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2, \tag{1.31}$$

the curves are ellipses, or circles after rescaling the axes. The origin is a fixed point. Every nonzero bounded orbit closes around it.

Closed and open curves can coexist in one phase space. A pendulum below the separatrix oscillates and traces a closed curve. Above the barrier it rotates continuously, producing an open branch when the angle is unwrapped. The same distinction appears for a particle moving through a periodic washboard potential.

Question from the audience. If the phase-space fluid is incompressible, how can neighboring energy contours bunch together and spread apart?

Response. Think of an incompressible fluid passing through a narrow channel. Where the channel narrows, the fluid moves faster; where it widens, the fluid moves more slowly. Closely spaced contours mean that the gradient of H is large, and Hamilton's equations therefore give a large phase-space speed. The density need not change.

Question from the audience. May closed and open constant-energy curves occur in the same system?

Response. Yes. A periodic potential provides the simplest example. Below a barrier, the particle is trapped in one well and the orbit is closed. Above it, the particle passes from well to well and the corresponding branch is open. The pendulum has the same oscillating and rotating regimes.

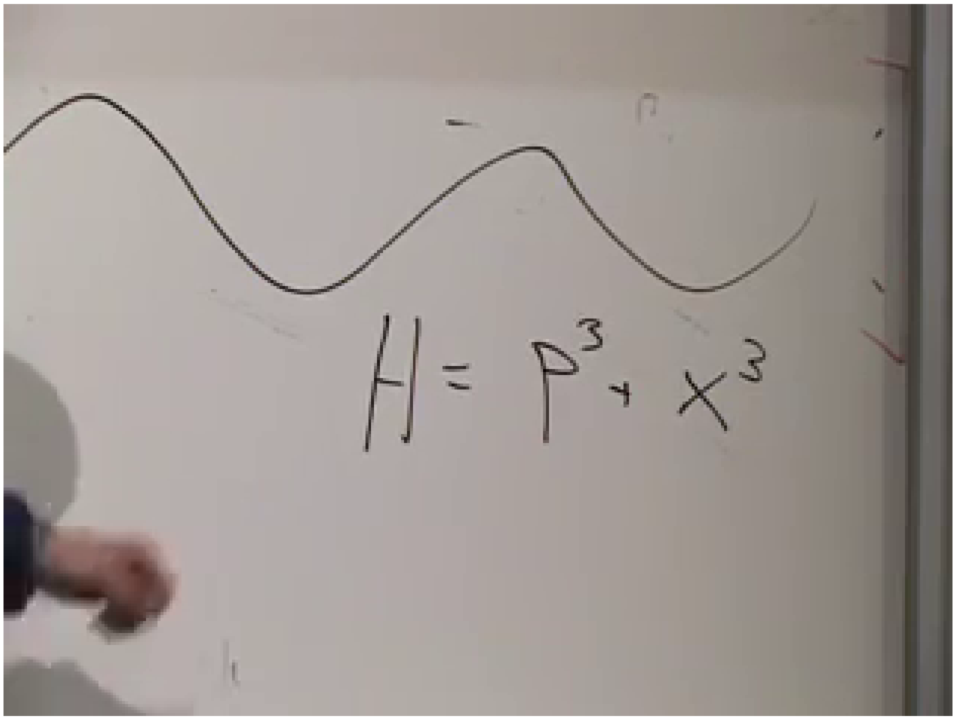

Figure 1.9: A deliberately nonstandard Hamiltonian at 01:25:50. It shows directly that the canonical momentum need not be proportional to velocity.

1.6.1 Momentum Need Not Be Mass Times Velocity

The relation $p = m\dot{q}$ belongs to the ordinary quadratic kinetic energy; it is not the definition of canonical momentum in general. Hamilton's first equation always says

$$\dot{q} = \frac{\partial H}{\partial p}. \tag{1.32}$$

For the illustrative Hamiltonian written in the lecture,

$$H = p^3 + q^3, \tag{1.33}$$

one finds

$$\dot{q} = 3p^2, \qquad \dot{p} = -3q^2. \tag{1.34}$$

Question from the audience. Is canonical momentum always a constant multiple of velocity?

Response. No. It is defined by $p_i = \partial L/\partial \dot{q}_i$, or equivalently related to velocity by $\dot{q}_i = \partial H/\partial p_i$. That relation may be nonlinear and may depend on the coordinates. The cubic example makes the nonlinearity obvious, although its momentum–velocity map is not globally one-to-one and therefore does not come from one globally regular ordinary Lagrangian.

Question from the audience. If microscopic Hamiltonian motion conserves energy, does that make every machine a perpetual-motion machine?

Response. No. Total energy conservation does not mean that all energy remains available as organized mechanical work. Energy can spread among enormous numbers of environmental degrees of freedom and become heat. Understanding that irreversibility and the distinction between total and usable energy requires statistical mechanics, entropy, and often chaotic dynamics.

1.7 Poisson Brackets and General Conservation Laws

Energy is not the only quantity that may be conserved. Let $A(q,p)$ be any function on phase space. Angular momentum in a plane is one example:

$$L_z = xp_y - yp_x. \tag{1.35}$$

For an observable with no explicit time dependence, the chain rule and Hamilton's equations give

$$\begin{aligned} \frac{dA}{dt} &= \sum_i \left(\frac{\partial A}{\partial q_i}\dot{q}_i + \frac{\partial A}{\partial p_i}\dot{p}_i \right) \\ &= \sum_i \left(\frac{\partial A}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial H}{\partial q_i} \right). \end{aligned} \tag{1.36}$$

This combination is the Poisson bracket. With the convention used here,

$$\boxed{\{A,B\} = \sum_i \left(\frac{\partial A}{\partial q_i}\frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i}\frac{\partial B}{\partial q_i} \right).} \tag{1.37}$$

Consequently,

$$\boxed{\dot{A} = \{A,H\}.} \tag{1.38}$$

The Hamiltonian is the generator of time evolution.

Taking $A = q_i$ or $A = p_i$ recovers Hamilton's equations:

$$\{q_i, H\} = \frac{\partial H}{\partial p_i} = \dot{q}_i, \qquad \{p_i, H\} = -\frac{\partial H}{\partial q_i} = \dot{p}_i. \tag{1.39}$$

The fundamental brackets are

$$\{q_i, q_j\} = 0, \qquad \{p_i, p_j\} = 0, \qquad \{q_i, p_j\} = \delta_{ij}. \tag{1.40}$$

A quantity $A(q,p)$ is conserved precisely when

$$\boxed{\{A,H\} = 0.} \tag{1.41}$$

For a rotationally invariant Hamiltonian, direct differentiation verifies $\{L_z, H\} = 0$. Later we will connect this calculation to symmetry; for now the bracket gives the mathematical test.

Question from the audience. How could Poisson have discovered such a combination of derivatives?

Response. Historical reconstruction is uncertain, but the combination is not arbitrary once Hamilton's equations are written. Apply the chain rule to an arbitrary $A(q,p)$, substitute the two canonical equations, and the bracket appears. Its importance is then confirmed by the algebraic properties and conservation laws it organizes.

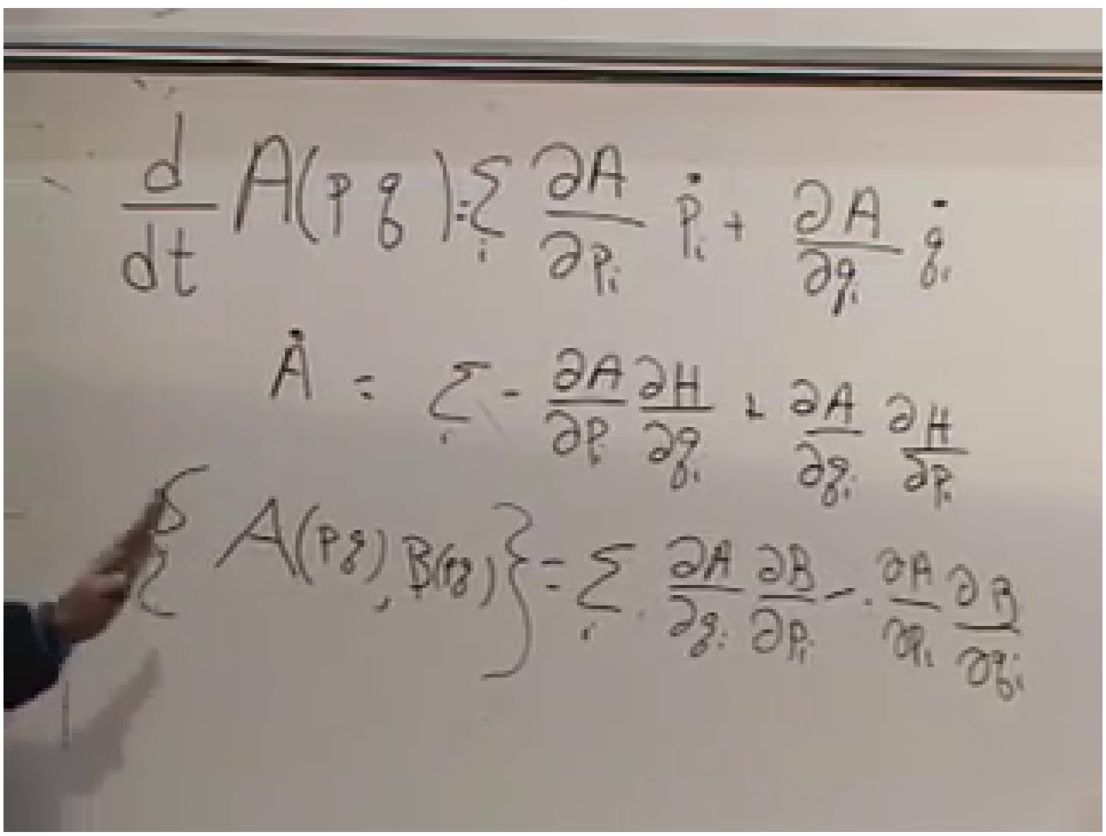

Figure 1.10: The chain rule and Poisson-bracket definition at 01:35:20. The bracket packages the crossed derivatives that already appeared in Hamilton's equations.

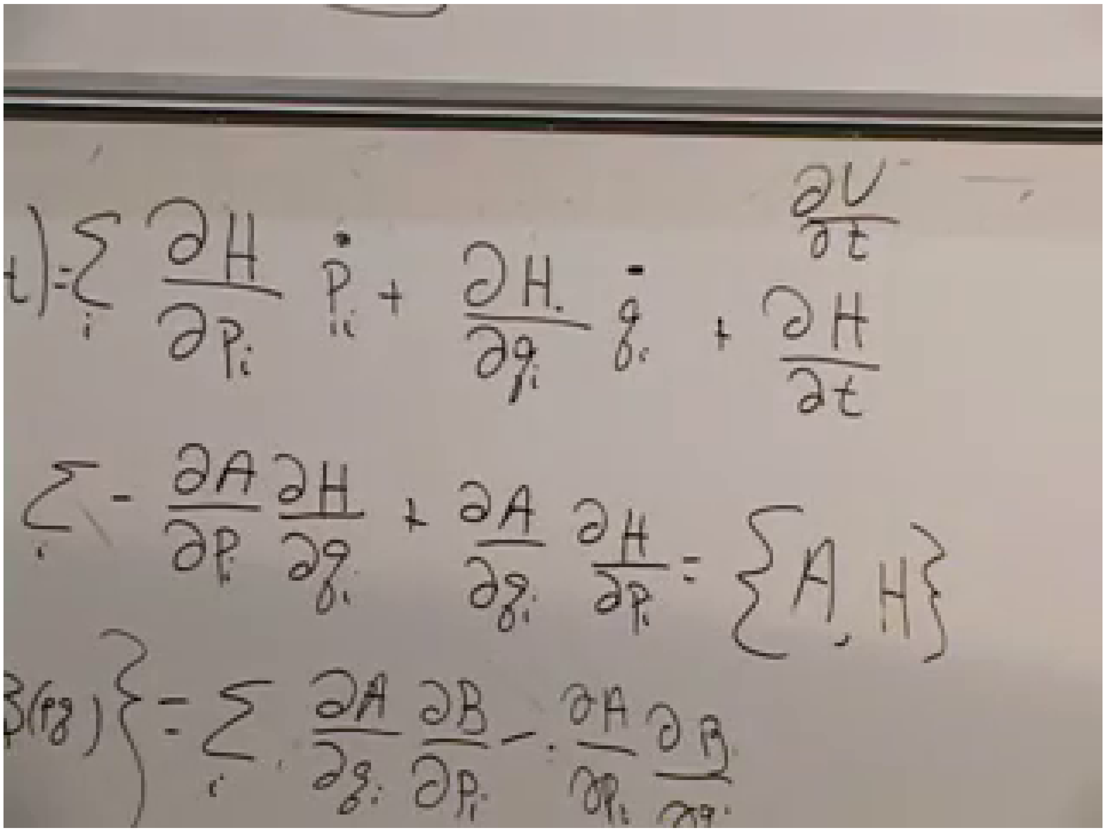

Figure 1.11: Explicit and implicit time dependence separated at 01:45:55. Hamilton's equations cancel the terms involving $\dot{q}_i$ and $\dot{p}_i$, leaving $dH/dt = \partial H/\partial t$.

1.8 Explicit Time Dependence

So far time entered only implicitly through the moving variables $q(t)$ and $p(t)$. An external apparatus can also make the Hamiltonian depend on time explicitly. A potential $U(q,t)$, for example, may be produced by a source whose prescribed position changes while the source itself is not included among the dynamical variables.

For a general observable $A(q,p,t)$,

$$\frac{dA}{dt} = \{A, H\} + \frac{\partial A}{\partial t}. \tag{1.42}$$

Setting $A = H$ gives

$$\frac{dH}{dt} = \{H, H\} + \frac{\partial H}{\partial t} = \boxed{\frac{\partial H}{\partial t}}, \tag{1.43}$$

because every function has zero Poisson bracket with itself. Thus a time-independent Hamiltonian is conserved, while explicit time dependence measures the energy supplied to or removed from the chosen subsystem.

The same conclusion follows directly from the time-dependent Legendre transform:

$$dH = \sum_i v_i\, dp_i - \sum_i \dot{p}_i\, dq_i - \frac{\partial L}{\partial t}\, dt, \tag{1.44}$$

so

$$\frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t} \tag{1.45}$$

when the canonical variables are held fixed on their respective sides of the transform.

Question from the audience. What is the difference between implicit and explicit time dependence?

Response. The value $H(q(t), p(t))$ changes its arguments as the system moves; that is implicit dependence. Explicit dependence remains even when q and p are held fixed, as in $U(q, t)$ produced by a moving external source. The canonical terms cancel in dH/dt; only the explicit partial derivative remains.

Question from the audience. Why do some books reverse the order or sign of the Poisson bracket? Which slot represents momentum?

Response. The names A and B label whole phase-space functions; neither argument slot is the momentum slot. The canonical variables q_i, p_i occur inside the definition. Since $\{A, B\} = -\{B, A\}$, reversing the order reverses the sign. Either convention is consistent if Hamilton's evolution equation is adjusted with it. Here we use $\dot{A} = \{A, H\}$.

Question from the audience. Is it the Poisson bracket that appears everywhere outside mechanics?

Response. The broad remark at this point concerns the Legendre transformation, which recurs throughout thermodynamics and other variational theories. Poisson brackets are fundamental in Hamiltonian mechanics and lead directly toward quantum commutators, but they are a different piece of the construction.

We began with a reversible map of abstract states and ended with a precise continuous law. The Legendre transformation changes variables from $(q, \dot{q})$ to (q, p); Hamilton's equations generate an incompressible phase-space flow; constant H confines that flow to energy surfaces; and the Poisson bracket tells how every observable changes. These are not separate tricks. They are four views of the same canonical structure.